Hardware Assembly & Integration Progress Report

Real-time 3-Phase Pump Power Abnormality Monitoring and Protection System

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1. Project Overview

This project implements a safety-critical embedded monitoring and protection system for Jahan Pump TYPE 4 series three-phase submersible drainage pumps (operating range 10–30 A). The system continuously measures voltage and current on all three phases, detects dangerous electrical abnormalities (over/under voltage, over-current, phase loss/imbalance), and automatically opens the existing contactor coil circuit to protect the motor from burnout. A manual reset is required after every trip. The work is developed as a professional open-source portfolio project for an Embedded Systems Master's degree application.

2. Hardware Platform

Category	Component	Specification
Controller	ESP32-WROOM-32	Dual-core, Wi-Fi/BT, 12-bit ADC, I2C
Precision ADC	ADS1115	16-bit, 4-channel, I2C (used for current)
Voltage Sensing	3 × ZMPT101B	Connected to ESP32 internal ADC pins
Current Sensing	3 × SCT-013 100A/50mA	33 Ω burden + bias → ADS1115
Actuation	5 V Relay Module	TONGLING JQC-3FF, opto-isolated
Display / UI	0.96" OLED + 3 LEDs + Button	SSD1306 + status indicators + manual reset
Power	ATX 5 V rail + 1 A fuse	Decoupled and fused low-voltage supply
Existing Panel	Hyundai HGC 25 + HMMS-32K	Contactor coil 220–240 V AC

3. Final Sensor Assignment (Updated)

To monitor all six signals simultaneously, the voltage sensors were moved from the ADS1115 to the ESP32 internal ADC. Current sensors remain on the higher-precision ADS1115.

Signal	Sensor	Connected To	Notes
Current Phase 1	SCT-013 + bias	ADS1115 AIN0	16-bit
Current Phase 2	SCT-013 + bias	ADS1115 AIN1	16-bit
Current Phase 3	SCT-013 + bias	ADS1115 AIN2	16-bit
Voltage Phase 1	ZMPT101B	ESP32 GPIO 35	12-bit ADC
Voltage Phase 2	ZMPT101B	ESP32 GPIO 32	12-bit ADC
Voltage Phase 3	ZMPT101B	ESP32 GPIO 33	12-bit ADC

4. Assembly Progress

Part	Description	Status
1	Power Supply (ATX 5 V + 1 A fuse)	Completed & Verified
2	Main Controller (ESP32 + OLED + LEDs + Relay + Button)	Completed & Tested
3	Current Sensors (3 × CT + burden + bias)	Completed & Verified
4	Voltage Sensors (3 × ZMPT101B)	Completed
5	ADS1115 + full 6-channel routing	Completed & Verified

5. Complete ESP32 Pin Map

ESP32 Pin	Function
VIN / 5V	Fused ATX +5 V power input
GND	Common ground
GPIO 21 (SDA)	I2C Data → OLED + ADS1115
GPIO 22 (SCL)	I2C Clock → OLED + ADS1115
GPIO 25	Relay control (contactor)
GPIO 26	Manual Reset button
GPIO 27	Red LED (Fault)

GPIO 14	Yellow LED (Warning)
GPIO 12	Green LED (Normal)
GPIO 35	Voltage Sensor 1 (ZMPT101B)
GPIO 32	Voltage Sensor 2 (ZMPT101B)
GPIO 33	Voltage Sensor 3 (ZMPT101B)
3.3V	OLED + current sensor bias circuits

6. Design Decision – Sensor Routing

The ADS1115 provides only four single-ended channels. To monitor all six signals (3 currents + 3 voltages) without adding a second ADC, the following trade-off was chosen:

- Current sensors remain on the ADS1115 (16-bit) because over-current protection requires higher accuracy and lower noise.
- Voltage sensors are read by the ESP32 internal 12-bit ADC. This is acceptable for detecting over/under voltage and phase loss.
- The main downside is higher noise and lower resolution on the voltage channels, which will be mitigated by software filtering and calibration.

7. Remaining Work

- Firmware: RMS calculation, protection logic, state machine, and trip handling
- Calibration of both current and voltage channels
- Safe wiring of the relay into the Hyundai HGC 25 contactor coil circuit
- Installation of CTs and voltage sensors on the live panel
- End-to-end testing on a real pump
- Final documentation and GitHub repository preparation

Report updated 28 September 2026 – reflects the decision to route voltage sensors to the ESP32 ADC so that all six measurement channels are active simultaneously.